6 **(a)** Define *magnetic flux*.

.....

..... [1]

(b) A circular coil of 300 turns and of radius 4.0 cm is placed in a uniform magnetic field of flux density 5.0×10^{-2} T. The coil is rotated in the magnetic field and the variation with time of magnetic flux linkage ϕ of the coil is sinusoidal as shown in Fig. 6.1.

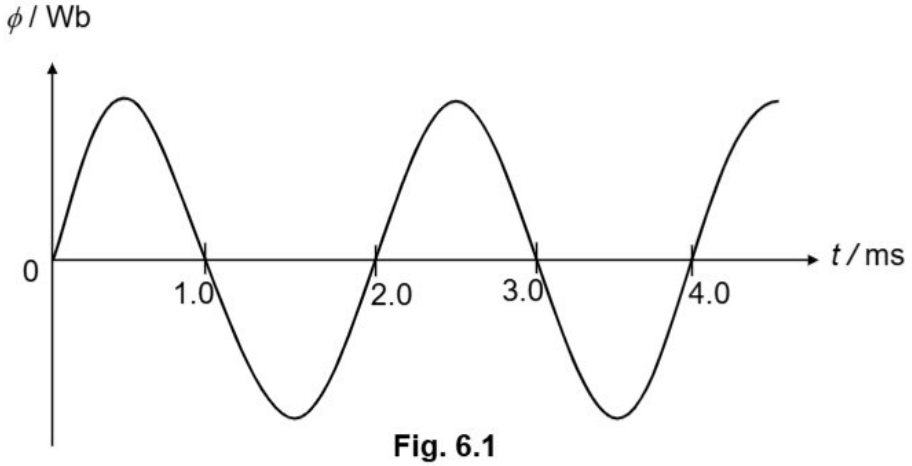


Fig. 6.1

(i) State the angle between the normal of the plane of the coil and the magnetic field at time = 0 s as shown in Fig. 6.1.

angle = ... ° [1]

(ii) Calculate the maximum magnetic flux linkage of the coil.

maximum magnetic flux linkage = ... Wb [2]

(iii) Calculate the average electromotive force (e.m.f.) induced across the coil when it rotates through the first quarter of a revolution.

average e.m.f. = ... V [2]

(iv) Explain how the minimum e.m.f. can be determined from Fig. 6.1.

State the first instant at which minimum e.m.f. is induced across the coil.

.....

.....

first instant of minimum e.m.f. = ... ms [2]

(v) Explain how the e.m.f. induced will vary as the coil rotates.

.....

.....

.....

..... [2]